COMMERCIAL ITEM DESCRIPTION

RESIN, ACRYLIC, DENTAL (Tray Material)

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Military Specification MIL-R-36433B which is cancelled.

This commercial item description covers a dental acrylic resin impression tray material, 1 lb.

Salient characteristics:

The acrylic resin shall be suitable for fabricating dental impression trays and baseplates.

Material. The acrylic resin shall consist of a liquid and a powder which when mixed in the proper proportion and manner (according to manufacturer's directions) shall produce a self-cured resin for forming dental trays and baseplates.

Liquid. The liquid shall be essentially an acrylic monomer with a suitable storage polymerization inhibitor. The liquid shall be clear, colorless, and substantially free from undissolved or particulate matter when examined without accessory magnification. This examination shall be made against a black background and against a white background each illuminated by a light providing an intensity of illumination which is no less than 100 foot-candles nor more than 350 foot-candles at a point 10 inches from its source.

Powder. The powder shall be essentially an acrylic polymer free from extraneous material such as dirt and lint. The powder may contain a maximum of 50 percent of a suitable inert filler provided the resin complies with the physical properties specified herein.

Toxicity. The manufacturer shall submit a statement that the mixing, the forming and use of the resulting dental tray shall have no toxic effect on a normal healthy person.

Color. The cured resin shall be a color which strongly contrasts with oral tissues so as to clearly delineate the outline of the impression tray when positioned intra-orally, and to identify the material from denture base materials. Color shall be uniformly distributed.

AMSC N/A FSC 6520

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Manufacturer's instructions. Instructions for storing and processing the resin shall accompany each package. Adequate and accurate instructions for using the resin in forming dental impression trays and baseplates, and in stabilizing baseplates shall also accompany each package. Two copies of instructions shall be forwarded to the contracting officer for information purposes. The instructions shall include the following minimum details:

a. Proper powder/liquid ratio.
b. Method of mixing, including time and temperature required.
c. Effect of temperature on mixing, working time, and procedures.
d. Method of forming impression trays, baseplates, and stabilizing baseplates.
e. Length of working time and setting time.
f. Method of trimming the cured resin to shape.

Particle size. The particle size shall be such that a minimum of 97.5 percent of the powder shall pass through a No. 80 U.S. Standard Sieve when tested as specified in the particle size test in the test section.

Working qualities of uncured resin. The uncured resin shall be plastic and workable for forming either an impression tray or a baseplate for a minimum of 5 minutes from start of mix, when tested, as specified in the working qualities test in the test section.

Working qualities of cured resin. The cured resin shall yield a satisfactory dental impression tray or baseplate, when processed according to the instructions furnished by the manufacturer and when tested. The cured resin shall lie inert or "dead" when applied to a cast without a spring back. The cured resin shall not exhibit graininess, cracks or pits and shall not be brittle.

Hardening time of uncured resin. The uncured resin shall harden and set within 12 minutes from start of mix when tested in accordance with ANSI/ADA No. 8 for Dental Zinc Phosphate Cement, time of setting, except that the relative humidity shall be 50%.

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Transverse deflection. The deflection of at least three of the five specimens shall be as specified in Table I when prepared and tested in accordance with the transverse deflection test of ANSI/ADA Specification No. 12 for Denture Base Polymers and as specified in test section under Transverse Deflection Test.

TABLE I. Deflection limits of specimens in transverse deflection test.

	DEFLECTION For loads (1) 1,500 grams (gm.) (3.31 pounds (lb.) to		
Load	2,500 gm. (5.51 lb.)	3,500 gm. (7.72 lb.)	
	Maximum millimeters (mm.)	Minimum mm.	Maximum mm.
Deflection	1.5 (0.059 inch) (in.)	1.0 (0.039 in.)	3.5 (0.138 in.)

(1) Deflections shall be measured from an initial load of 1500 gm. (3.31 lb.) to the load indicated.

Unit size. The amount of powder and liquid supplied as a unit, when mixed as directed, shall be sufficient to make a minimum of 22 mixes of at least 20 cubic centimeters each of cured resin, when tested as specified in the unit size test in the test section.

Delivery schedule. Not more than four months shall have elapsed from the date of manufacture of the resin to the date of delivery to the Government.

Lot or control number. Each unit of issue shall be marked with the lot or control number. Each lot or control number shall consist of a maximum of 1500 unit packages of the acrylic resin.

Accessories. A suitable powder scoop and liquid measure shall be included in each unit of issue. In addition, if the resin is formulated for pouring consistency, a tray former shall be supplied.

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Workmanship. The acrylic resin and its containers shall be free from defects which either detract from their appearance or may impair their serviceability.

Unit. Package (PG). One package, consisting of one container of powder resin having a minimum net weight of one pound, and one container of liquid containing 8 fluid ounces (minimum), as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 1.5 and an inspection level of II. Unit of product for examination purposes shall be one unit of issue of acrylic resin.

Sampling for tests. Sampling for tests shall be conducted as specified in table II. Acceptance number shall be zero.

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TABLE II. Tests.

Characteristic	Number of determinations per lot or batch number	Unit of product
Particle size	1	1 container of powder
Working qualities: of uncured resin	1	1 unit package
of cured resin	1	1 unit package
Hardening time	1	1 unit package
Transverse deflection	1 (1)	1 container of liquid
Unit size	2 (2)	1 unit package

(1) Each determination shall be the average of 5 specimens.

(2) Two units of product to be tested per lot or batch number; a lot shall be rejected if any determination fails the test requirement.

Examination. Acrylic resin shall be examined for defects including, but not limited to those listed below.

Defects.

Unit package not complete with powder and liquid.
Dirt or lint present in powder.
Liquid not free from undissolved or particulate matter.
Liquid not clear or colorless.
Manufacturer's instructions not present in unit package; not complete (see manufacturer's instructions).

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Closure for bottles of powder not as specified.
Closure for bottles of liquid not as specified.
Powder scoop and liquid measure not present in unit package.
Packaging, packing, and marking not as specified.

Tests.

Particle size. Particle size of the powder shall be determined by taking ten to twelve grams of the resin powder and placing it on a number 80 sieve (180 um) conforming to ASTM-E11. The top and bottom shall be replaced, and the assembly shall be vibrated mechanically in a Ro-tap for 2 minutes. On completion of the test, the sieve shall be carefully examined for residue remaining on the upper surface of the wire or occluding the sieve openings.

Working qualities test. The resin shall be prepared according to manufacturer's instructions at 23.0°C. ± 2.0°C. (73.4°F. ± 3.6°F.). The mix shall be used to make a dental impression tray on an endentulous plaster or stone model in accordance with normal dental prosthetic techniques. The resin shall be tested for the working qualities specified in working qualities of uncured resin, working qualities of cured resin and color.

Transverse deflection test. A resin mix shall be prepared in accordance with manufacturer's instructions at 23.0°C ± 2.0°C. (73.4°F ± 3.6°F) in a sufficient size to yield five specimens. The final preparation and testing shall be in accordance with ANSI/ADA Specification No. 12 for Denture Base Polymers except that Table III, as specified herein, shall be used for the schedule of loading and reading.

TABLE III. Schedule of loading and reading.

Time		Load	Dial Reading (1)
Min.	Sec.	gm.	mm.
0	00	1,500	
0	30	1,500-----	---------A
1	00	2,000	
1	30	2,000	
2	00	2,500	
2	30	2,500-----	---------B
3	00	3,000	
3	30	3,000	
4	00	3,500	
4	30	3,500-----	---------C

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Deflection at 2,500 gm. = B minus A
Deflection at 3,500 gm. = C minus A
(1) Readings shall be made to the nearest 0.01 mm.

Unit size test. A portion of powder and liquid shall be mixed according to manufacturer's instructions, and the volume shall be calculated to the nearest cubic centimeter. Calculate the number of mixes (20 cubic centimeters minimum each) obtainable from unit package.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

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Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document.

National Stock Number	Item Identification
6520-00-889-9566	RESIN, ACRYLIC, DENTAL, IMPRESSION TRAY, 1 lb.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:
Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:
VA-OSS
PHS
FDA-MPQAS

PROJECT NO. 6520-2070

LOCATION: NCR DISK # 5

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-53604	RESIN, ACRYLIC, DENTAL (Tray Material)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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